

Key question for an operator T on a finite dim'l v-space V over field F .

How to find a good basis for V wrt which T has a nice matrix?

Nice = diagonal, if possible Subquestions

In this case we say " T is diagonalizable"

ie $\exists$ a basis wrt which the matrix of T is diagonal

ie $\exists$ a basis $B = (v_1, v_2, \dots, v_n)$ of V such that

$$T(v_1) = \lambda_1 v_1, \dots, T(v_n) = \lambda_n v_n$$

Matrix of T wrt $B = \text{diag}(\lambda_1, \dots, \lambda_n) =: D$

When $V = F^n$
 $T(x) = Ax$ for
 $n \times n$ matrix A

We want a basis of F^n consisting of n column vectors $v_1, \dots, v_n$ st. $Av_i = \lambda_i v_i \quad \forall i=1, \dots, n$

let $P = [v_1 \dots v_n]$ then we have

$$P^{-1}AP = D \iff AP = PD \iff (*)$$

$$AP = A[v_1 \dots v_n] = [Av_1, \dots, Av_n]$$

$$PD = [v_1 \dots v_n] \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix} = [\lambda_1 v_1, \dots, \lambda_n v_n]$$

So (*) again gives $Av_i = \lambda_i v_i \quad \forall i$

Thus to diagonalize T , we need to solve the equation $T(v) = \lambda v$ for both λ and nonzero v (and hope that

we get a basis of V consisting of such magic vectors)

Def: For $\lambda \in F$,
 λ -eigenspace

of T is $\ker(T - \lambda I)$

Defⁿ Given an operator T on v.space V , we say $v \in V$ is an eigenvector of T if $v \neq 0$ and $T(v) = \lambda v$ for $\lambda \in F$

Then λ is called $\left\{ \begin{array}{l} \text{the eigenvalue of } T \\ \text{corresponding to eigenvector } v \end{array} \right.$

We will first find eigenvalues of T and then the corresponding eigenvectors. How?

$$\begin{array}{l|l} Tv = \lambda v & \Leftrightarrow (T - \lambda I)v = 0 \rightarrow \text{Note: } \lambda v = \lambda \text{Id}(v) \\ \Leftrightarrow Tv - \lambda v = 0 & \Leftrightarrow v \in \ker(T - \lambda I) = \ker(\lambda I - T) \end{array}$$

Upshot λ is an eval (= eigenvalue)
for $T \iff \ker(T - \lambda I) \neq \{0\}$

$\iff T - \lambda I$ is not invertible

$$\iff \boxed{\det(T - \lambda I) = 0}$$

So to find evals of T , treat λ as an unknown scalar. Get a matrix for $T - \lambda I$. Calculate its det. Equate to 0. Solve for λ .

Exercise: Espaces corresponding to distinct evals are linearly independent! Redo oblique reflⁿ example!

Examples ① Fibonacci matrix

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \quad A - \lambda I = \begin{bmatrix} 1-\lambda & 1 \\ 1 & -\lambda \end{bmatrix}$$

$$\det \begin{pmatrix} 1-\lambda & 1 \\ 1 & -\lambda \end{pmatrix} = -\lambda(1-\lambda) - 1 = \lambda^2 - \lambda - 1$$

Roots = evals of A are $\lambda_i = \frac{1 \pm \sqrt{5}}{2}$ ($i=1,2$)

Find corresponding evectors by calculating

$$\ker(T - \lambda_1 I) \quad \ker(T - \lambda_2 I)$$

$$\psi \\ v_1 \neq 0$$

$$\psi \\ v_2 \neq 0$$

Defⁿ For an operator T on V
characteristic polynomial of T
is $\det(\lambda I - T)$ regarded as
a polynomial in variable λ
"eigenpolynomial"?

Example 2 $V = \mathbb{R}^2$, $T = \text{rotation by } \frac{\pi}{2}$

$$\text{matrix of } T = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

$$\text{char poly} = \det \begin{bmatrix} \lambda & 1 \\ -1 & \lambda \end{bmatrix} = \lambda^2 + 1$$

No root in $\mathbb{R}$ as expected!

Rmk If I regard $T(v) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} v$

for $v \in \mathbb{C}^2$ then T will have
eigenvalues $i, -i$ and in fact we can diagonalize T !

Do it. Calculate $\ker(T - iI) \ni w_1 \neq 0$

$$\ker(T + iI) \ni w_2 \neq 0$$

w_1, w_2 being eigenvectors with distinct eigenvalues cannot
be scalar multiples of each other. So they must form
a basis of $\mathbb{C}^2$.